

Rowan Torbitzky-Lane

rowan.a.tl@protonmail.com • rowantl.github.io • github.com/RowanTL • linkedin.com/in/RowanTL

Education

B.S. Computer Science - Minor in Applied Mathematics - GPA: 3.80 Missouri University of Science and Technology December 2024
B.S. Mathematics - GPA: 4.00 Lindenwood University May 2027

Purpose

To work at Revel as a Software Engineer - Test. Python developer by training, self-taught and test-driven NixOS and Rust programmer. Highly motivated and loves to learn by experience.

Skills

- | | | |
|----------|---------|--------|
| • Python | • CI/CD | • CUDA |
| • Rust | • SQL | • Git |
| • Docker | • Linux | • Nix |

Experience

Computer Vision Intern - *USC Institute for Creative Technologies* May 2026 - August 2026

- Leverage Rust FFI to calculate mesh/splat correspondences in under one second
- Train Gaussian Splats on bounded and unbounded scenes
- Extract single objects from Gaussian Splats
- Utilize Nix for programming environment setup

NSF Research Assistant - *Missouri University of Science and Technology* May 2024 - October 2024

- Train novel, bi-directional LLMs on Genome Sequences to correct errors in genome reads
- Clean and analyze textual data with Scikit-learn and Scikit-bio
- Incorporate huggingface library for model testing

Research Assistant - *Center for Intelligent Infrastructure* May 2023 - October 2023

- Calculate Optical Flow with MATLAB and C++ utilizing Slurm on campus HPC
- Coordinate Web Requests with Django to interact with internal PostgreSQL server
- Improve frontend with Typescript, such as adding video upload capability
- Extract image features using OpenCV such as utilizing optical flow to predict annotations in subsequent video frames
- Perform Annotations for image segmentation Convolutional Neural Networks

Data Science Co-Op - *Hunter Engineering Company* January 2022 - August 2022

- Utilize YOLO and PyTorch in license plate recognition problems
- Pioneer object detection models to eradicate bad data before annotators see it
- Innovate data pipeline by automating our data downloading process
- Design and containerize APIs with FastAPI and Flask within Docker, used to create endpoints within Azure
- Leverage Pandas and Microsoft SQL in data processing pipelines
- Implement syntax formatting actions into internal Azure DevOps git repository ensuring up to par formatting
- Utilize Excel for data annotation involving writing macros to automate data exportation
- Transition to storing data in AWS S3 Bucket Storage

Personal Projects

Self-Hosted NixOS Server - *SQL, YOLO, Docker, Firewalld, Git, Nginx, CUDA* January 2023 - Present

- Automate flake updates daily via CI/CD pipeline on GitLab
- Train bird object detection Convolutional Neural Networks
- Host game servers via Docker
- Point domain names via A records

Rust Genetic Programming Framework - *Polars, rust_decimal* March 2025 - Present

- Perform pre-processing on Time Series Data
- Implement test driven development philosophy involving writing unit tests
- Write declarative and procedural macros for reduced boilerplate
- Utilize rust_decimal's fixed-point precision for increased accuracy

Talks & Events

Talk: Genetic Programming in Python @ PySTL https://www.youtube.com/watch?v=ZzYi_j2IYuY September 18th, 2025

Talk: Actix Web Framework @ STL Rust <https://rumble.com/v72mid6-actix-web-unleashed.html> December 4th, 2025

Hackathon: SIGGRAPH 2026 Worlds in Action Hack @ MG Studio <https://luma.com/u7pmhw92> July 18th, 2026